

Indoor Air Quality Dataset of Public Elementary Schools in the San Luis Potosí Metropolitan Area

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Abstract—

This paper presents one of the first experimental indoor air quality (IAQ) datasets collected in the San Luis Potosí Metropolitan Area, a dusty semi-arid region characterized by rapid urban growth and industrial activity. The dataset, distributed in a semicolon-separated text format, was acquired during a longitudinal monitoring campaign conducted between October 2023 and June 2024 in public elementary schools exposed to multiple IAQ-related environmental stressors. The dataset contains 174260 records collected at an average sampling interval of 10 minutes using an Internet of Things (IoT)-based monitoring infrastructure. Each record includes: (i) the context information including the timestamp and the identifier of the sensing node associated with the monitored classroom; (ii) indoor particulate matter concentration, expressed as $PM_{2.5}$ mass concentration ($\mu g/m^3$); (iii) carbon dioxide (CO_2) concentration; (iv) complementary environmental variables, namely air temperature and relative humidity; and (v) Pupil Health Air Quality Index (PHAQI). Beyond the dataset itself, this work provides the associated source code, metadata, and contextual information required to support data preprocessing, organization, and analysis. The accompanying resources also enable researchers to reproduce the experimental workflow and generate customized synthetic datasets for benchmarking, comparative studies, and future IAQ research applications.

I. INTRODUCTION

Studying indoor air quality (IAQ) in schools is crucial. IAQ refers to the condition of air within enclosed spaces and directly affects human comfort, cognitive performance, and health outcomes, particularly among sensitive populations like children and elderly individuals [1]. This quality directly impacts pupils and students' health and concentration. Schools are small spaces where many people spend numerous hours each day, making them highly susceptible to poor air quality. For this reason, several datasets, based on IAQ in the context of schools, have been proposed by research community in the literature. As examples, we can cite the three datasets of *SINPHONIE project* (Schools Indoor Pollution and Health: Observatory Network in Europe) [2], *EDIAQI* (Evidence Driven Indoor Air Quality Improvement) [3] and Stellenbosch (South Africa) educational environments [4]. However, no experimental dataset collected in a dusty semi-arid region like Mexico cities exist. For this reason, we present in this paper, one of the

first indoor air quality (IAQ) experimental dataset (organized in a semicolon separated text format) collected from San Luis Potosí Metropolitan Area. The dataset comprise context information (timestamp and classroom identifier) in addition to indoor environmental parameters (relative humidity (RH), carbon dioxide (CO_2), particulate matter ($PM_{2.5}$)) acquired under normal classroom operating conditions, together with health level labels derived from the proposed indicator.

The remainder of this paper is structured as follows. Section II outlines the methodology employed for data acquisition, preprocessing, and dataset construction. Section III presents the formulation of the Personalised Hybrid Air Quality Index (PHAQI). Section IV provides a detailed characterization of the resulting indoor air quality (IAQ) dataset through descriptive and multivariate analyses, including Pearson correlation analysis, Principal Component Analysis (PCA), and the distribution of PHAQI classes.

II. METHOD

A. Context and location

To construct the IAQ dataset, a field study was conducted across 53 elementary schools located in San Luis Potosí, Mexico. The aim was to gather empirical indoor environmental data representative of real classroom conditions. The data collection focused on four key parameters known to influence indoor air quality: carbon dioxide (CO_2) concentration, fine particulate matter ($PM_{2.5}$), relative humidity, and temperature. These parameters were continuously monitored using environmental sensors deployed in classrooms, resulting in a comprehensive dataset that captures spatial and temporal variations in IAQ across diverse educational settings within the San Luis Potosí Metropolitan Area (SLPMA). The data collection process was carried out in two main steps to support the analysis of Indoor Air Quality (IAQ) in the context of elementary schools in Mexico City.

The first step involved collecting real-world data from elementary schools. IAQ monitoring was conducted in 11 schools selected from our sample, as detailed in the *Methods* section. In each school, one representative classroom was selected for environmental monitoring.

Subject	Data Science / Environmental Monitoring / Indoor Air Quality
Specific subject area	Experimental Indoor Air Quality (IAQ) dataset collected using an IoT-based monitoring infrastructure in public elementary schools located in the San Luis Potosí Metropolitan Area, Mexico. The dataset includes time stamp, classroom identifier, $PM_{2.5}$ concentration, CO_2 concentration, air temperature, and relative humidity measurements.
Data format	Raw; Processed
Type of data	Table; Semicolon-separated text files (.csv)
Data collection	The experimental data collection campaign was conducted between October 2023 and June 2024 in public elementary schools situated in the San Luis Potosí Metropolitan Area, Mexico. Indoor air parameters were monitored using an IoT-based sensing infrastructure deployed inside classrooms. Measurements were collected at an average sampling interval of 10 minutes. Each IoT sensing node continuously monitored particulate matter concentration ($PM_{2.5}$), carbon dioxide (CO_2), air temperature, and relative humidity. The collected measurements were automatically validated, timestamped, and stored for subsequent preprocessing, sorting, and analysis.
Data source location	10 classrooms in public elementary schools, San Luis Potosí Metropolitan Area, San Luis Potosí, Mexico.
Data accessibility	Geographic context: Semi-arid urban–industrial environment. Repository name: [To be completed after repository publication] Data identification number: DOI: [To be assigned] Direct URL to dataset: [Insert repository URL]

The timing and scheduling of the monitoring sessions were coordinated with school authorities, teachers, and students to ensure minimal disruption to regular classroom activities. Data collection took place during epidemiological weeks 23 and 48 of the year 2023, representing the winter and spring seasons, respectively.

B. Calibration and measurements

To ensure the reliability of air quality measurements in our study, we calibrated both CO_2 and $PM_{2.5}$ sensors prior to deployment across participating classrooms. For CO_2 sensors, which utilized non-dispersive infrared (NDIR) technology, calibration was conducted using a two-point reference procedure. Each device was first exposed to a zero-gas environment, specifically high-purity nitrogen, to establish the baseline reading. This was followed by exposure to a certified gas mixture containing a known concentration of CO_2 (typically 1000 ppm), allowing for span adjustment. This procedure established a calibration curve for each sensor, which was then validated against ambient outdoor air concentrations assumed to be near 400 ppm. In classrooms where automatic baseline correction algorithms were active, we reviewed sensor placement to ensure periodic exposure to fresh air, avoiding artificially elevated baselines due to continuous occupancy (Cai et al., 2024)[XX]. For $PM_{2.5}$ sensors, calibration was conducted via co-location with a reference-grade gravimetric instrument in a controlled indoor setting over a two-week period. Sensors recorded optical particle counts in parallel with mass concentrations obtained from the reference device. These datasets were then analyzed to generate a correction factor using regression modeling, accounting for temperature and relative humidity, both of which influence particle scattering and sensor accuracy. Calibration models were embedded in the device firmware prior to field deployment. Additionally, we performed seasonal calibration

validation to account for variations in aerosol composition and ambient moisture, which are characteristic of the semi-arid conditions in San Luis Potosí (Park et al., 2021)[XX]. Temperature and relative sensors were also calibrated using standard techniques.

The IAQ parameters collected included carbon dioxide (CO_2) concentration, particulate matter ($PM_{2.5}$), temperature, and relative humidity. These were recorded at one-minute intervals throughout regular school hours (8:00 a.m. to 2:00 p.m.) using pre-programmed IoT monitoring devices. This setup enabled continuous, real-time monitoring of indoor environmental conditions, providing a detailed view of the air quality experienced by both students and staff during daily academic activities. Although data collection spanned the entire day, classroom activities occurred between 8:00 a.m. and 2:00 p.m., Monday through Friday. Each monitor was wirelessly connected—via Bluetooth or Wi-Fi—to the Air Sentinel platform described in the following section, enabling real-time data synchronization, centralized access, and secure storage.

C. Data processing

In this part, we describe the procedures followed to process the raw data collected during the IAQ monitoring campaign, as introduced previous phase. The main objective of the data preparation phase double. On one hand, is to clean, structure, and enrich the data to achieve a cleaned dataset. On the second hand, use this database to create a supervised machine learning (ML) model that predicts indoor air quality, ensuring consistency and comparability between multiple classes and schools.

The raw dataset was initially stored in a private cloud and consisted of 13 CSV files, each corresponding to an individual classroom in Mexico City. These files contained time-stamped readings for four environmental variables: CO_2

U.S. Environmental Protection Agency ASHRAE comfort recommendation WHO guidance			Pupil Health Air Quality Index (PHAQI)	
Excellent	Excellent indoor conditions with no expected health.	1		
Acceptable	Acceptable conditions; routine monitoring is recommended	2		
Degraded	Degraded air quality that may cause discomfort, particularly for sensitive-pupils.	3		
Unhealthy	Unhealthy conditions requiring corrective actions, such as improved ventilation or reduced occupancy	4		
Very unhealthy	Very unhealthy environment associated with significant health risks for pupils.	5		
Hazardous	Hazardous conditions requiring immediate intervention, such as enhanced ventilation or temporary evacuation	6		

Fig. 1. PHAQI index interpretation

concentration, $PM_{2.5}$ levels, temperature, and relative humidity. After downloading the files, exploratory data analysis (EDA) was conducted using Jupyter Notebooks to understand the data structure, verify its integrity, and identify potential anomalies. The 14 files were then merged into a single consolidated dataset to facilitate further analysis. A new column, “Sensor_ID,” was added to differentiate data from each classroom, using the unique MAC address of the respective IoT sensor (e.g., D4DA5CC856). The resulting dataset contained approximately 286,000 lines of data collected between October 31, 2023, and June 28, 2024, covering both winter and spring seasons.

Subsequently, we applied a series of cleaning operations to address issues such as missing values, inconsistent timestamps, and outliers. Timestamps were normalized to a standard timezone and formatted using ISO-8601 to ensure consistency. Missing values were imputed using linear interpolation when possible, while non-physical or extreme values (e.g., CO_2 above 5000 ppm or negative $PM_{2.5}$) were either corrected or removed based on domain thresholds and statistical outlier detection (e.g., z-score analysis). Feature engineering was then performed to enhance the dataset with temporal indicators such as hour of day, day of week, and season tags, which capture daily and seasonal IAQ patterns. Additional features were derived from the raw values, including rolling averages and first-order differences, which help in identifying trends and abrupt variations. Furthermore, an indirect occupancy proxy was computed based on CO_2 rate-of-change, enabling a better understanding of classroom activity levels without intrusive methods.

III. PUPIL HEALTH AIR QUALITY INDEX

To enable a health-oriented and machine-learning-ready representation of indoor air quality in educational environments, we propose a composite indicator, denoted as *PHAQI*, which integrates ventilation-related risk and particulate pollution exposure. This indicator is derived by combining carbon dioxide

(CO_2) concentration thresholds defined by ASHRAE¹ [5] with particulate matter ($PM_{2.5}$) exposure levels expressed through the U.S. Environmental Protection Agency (EPA) Air Quality Index (AQI) [6]. The algorithm (1) represents how it is built.

Algorithm 1 Derivation of *PHAQI* Using 5-Level ASHRAE and 6-Level EPA AQI

Require: CO_2 concentration (ppm), $PM_{2.5}$ concentration ($\mu g/m^3$)

Ensure: $C_{PHAQI} \in \{1, \dots, 6\}$

- 1: Determine ASHRAE class $C_{CO_2} \in \{1, \dots, 5\}$
- 2: Map C_{CO_2} to unified risk level $f(C_{CO_2}) \in \{1, \dots, 6\}$
- 3: Compute AQI from $PM_{2.5}$ using EPA breakpoints
- 4: Determine AQI class $C_{AQI} \in \{1, \dots, 6\}$
- 5: $C_{PHAQI} \leftarrow \max(f(C_{CO_2}), C_{AQI})$
- 6: **return** $C_{PHAQI} = 0$

Formally, EPA AQI is computed using “24-hour” averages for $PM_{2.5}$. However, for real-time IAQ monitoring and ML-based prediction, it is common (and acceptable in research) to compute “instantaneous AQI” using the same breakpoints as presented in the Figure ().

In the proposed approach to calculate *PHAQI*, each ASHRAE CO_2 class is conservatively mapped onto the six-level AQI scale to preserve health-risk severity. The resulting ASHRAE-derived risk level and the $PM_{2.5}$ -based AQI class are then aggregated using a worst-case rule, whereby the final *PHAQI* class is defined as the maximum of the two as presented in the algorithm (1). This strategy ensures that any critical exposure—whether due to insufficient ventilation or elevated particulate pollution—dominates the final health assessment, thereby favoring a precautionary and health-protective interpretation. The interpretation of the final “*PHAQI*” levels is summarized as follows in Figure (1).

¹<https://www.co2meter.com/fr-fr/blogs/news/ashrae-co2-standards-classrooms>

IV. DATASET ANALYSIS

A. Dataset Composition of IAQ parameters

The dataset used in this study is publicly available in [?] and was specifically collected to investigate the impact of key IAQ parameters - namely CO_2 , $\text{PM}_{2.5}$, humidity, and temperature - on pupils' health in Mexican elementary schools. These parameters were retained for their direct relevance to children's health and learning conditions, and for their recognition by the IAQ community through established recommendations and international standards. Table I summarizes the environmental and contextual parameters considered in this study.

TABLE I
SUMMARY OF INDOOR AIR QUALITY (IAQ) PARAMETERS COLLECTED

Parameter	Unit	Description
Timestamp	Date-Time	Measurement date and time recorded in the format DD/MM/YYYY HH:MM:SS (e.g., 31/10/2024 10:20:07).
ID class	Categorical	Anonymized IAQ class label (A, B, C, D) derived from IAQ standards and used for supervised machine learning.
CO_2 Concentration	ppm	Indicates ventilation efficiency and occupant density; high levels are associated with drowsiness and decreased cognitive performance.
$\text{PM}_{2.5}$	$\mu\text{g}/\text{m}^3$	Fine particulate matter with diameter ≤ 2.5 μm ; associated with respiratory and cardiovascular risks, especially in children.
Temperature	$^{\circ}\text{C}$	Monitored to ensure thermal comfort and assess how it interacts with other IAQ factors.
Humidity	%	Affects comfort and the dispersion of airborne pollutants; both too high and too low levels can promote health issues.
PHAQI	Health level	Indicates the Pupil Health Air Quality Index.

At the end, the resulting dataset contained 174260 samples described by 6 parameters. These include four environmental variables, namely $\text{PM}_{2.5}$, CO_2 , temperature (T), and humidity (H), as well as two spatio-temporal variables: classroom location identifier IDclass and *timestamp*.

To better understand the relationships among the indoor air quality (IAQ) parameters, a Pearson correlation analysis was conducted, as illustrated in Table II.

TABLE II
PEARSON CORRELATION MATRIX OF INDOOR AIR QUALITY PARAMETERS

	ID class	CO_2	$\text{PM}_{2.5}$	Humidity	Temperature	PHAQI
ID class	1.00					
CO_2	0.06	1.00				
$\text{PM}_{2.5}$	0.01	-0.00	1.00			
Humidity	-0.34	0.11	0.01	1.00		
Temperature	0.34	0.02	0.00	-0.71	1.00	
PHAQI	0.33	0.07	0.02	-0.78	0.80	1.00

The strongest positive correlation was observed between temperature and PHAQI ($r = 0.80$), indicating that thermal conditions significantly influence the estimated health-related air quality level. This suggests that increasing temperature

values are associated with higher PHAQI scores during the experimentation period.

Conversely, humidity exhibited a strong negative correlation with PHAQI ($r = -0.78$), implying that elevated humidity levels may degrade perceived indoor air quality and potentially affect occupants' comfort and health conditions. In addition, temperature and humidity showed a substantial inverse relationship ($r = -0.71$), confirming the expected environmental dependency between these two parameters in indoor spaces.

The correlations involving CO_2 concentration and $\text{PM}_{2.5}$ remained weak across most variables. CO_2 showed only minor positive correlations with humidity ($r = 0.11$) and PHAQI ($r = 0.07$), while $\text{PM}_{2.5}$ demonstrated nearly negligible correlations with all other parameters. These findings suggest that, within the considered experimental environment, thermal comfort indicators had a greater influence on the global IAQ assessment than particulate matter or ventilation-related indicators.

Regarding the anonymized IAQ class labels, moderate correlations were observed with temperature ($r = 0.34$) and PHAQI ($r = 0.33$), whereas humidity presented a moderate negative correlation ($r = -0.34$). This indicates that the IAQ classification mechanism is primarily influenced by thermal and comfort-related factors.

B. PCA-Based Feature Analysis

The Principal Component Analysis (PCA) was applied as an exploratory step to investigate the underlying structure of the IAQ dataset and to identify the most influential parameters prior to ML model training. The resulting loadings for the two first principal components are reported in Figure 2.

	PC1	PC2
Idclass	0.449403	0.259282
co2	-0.032988	0.952640
pm25	0.003242	0.033534
humidity	-0.633837	0.154057
temperature	0.628640	0.019792

Fig. 2. PHAQI index interpretation

The first principal component (PC1) is primarily driven by temperature and humidity, highlighting the dominant influence of thermo-hygrometric conditions on indoor environmental variability. Notably, the opposite signs observed for these two parameters indicate a strong negative correlation between temperature and humidity, which is physically consistent with typical indoor climate behavior. The second principal component (PC2) is overwhelmingly dominated by CO_2 concentration, suggesting that ventilation and occupancy-related effects constitute an independent and distinct source of variability within the dataset. Conversely, $\text{PM}_{2.5}$ exhibits very low loadings across both principal components, indicating a limited contribution to the overall variance structure of the analyzed IAQ dataset.

According to the PCA results, the analyzed IAQ variables can be separated into two distinct groups: *dominant variables*, represented by temperature and relative humidity, which account for the highest proportion of variance within the dataset, and *secondary variables*, represented by CO₂ and PM_{2.5}, whose contribution to data discrimination is comparatively lower.

These findings emphasize the importance of considering climatic parameters when evaluating indoor air quality in dusty semi-arid regions, such as urban areas in Mexico. In contrast to numerous studies that primarily focus on CO₂ and particulate matter indicators, our analysis demonstrates that excluding temperature and humidity may result in an incomplete representation of IAQ dynamics. Consequently, the integration of thermo-hygrometric variables appears fundamental for robust and context-sensitive IAQ assessment in semi-arid environments.

C. Overall Distribution of Indoor Air Quality Classes

Prior to conducting the statistical and predictive analyses, it is important to examine the structure and representativeness of the collected dataset. In this perspective, we introduce the Figure (3), to illustrate descriptive visualizations to characterize the distribution of PHAQI index (indoor air quality classes) both at the global dataset level and within individual monitoring groups.

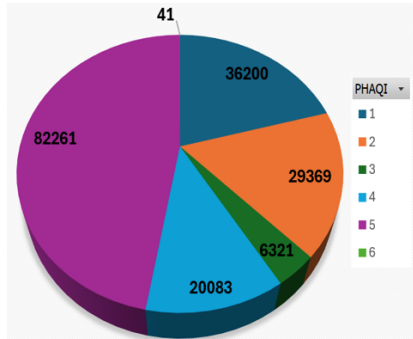


Fig. 3. PHAQI levels distribution

Figure (3) illustrates the overall distribution of indoor air quality (IAQ) classes in the complete dataset. The figure was generated by aggregating all observations across the monitoring groups and calculating the frequency of each IAQ class. The resulting counts were visualized using a pie chart, where each sector represents the proportion of observations belonging to a specific IAQ category. This representation provides a global overview of the dataset composition and highlights the relative prevalence of the different IAQ conditions.

The main objective of this figure is to assess the overall class balance of the dataset before conducting further statistical analyses or developing predictive models. Understanding the distribution of target classes is particularly important in data-driven studies because severe class imbalances may influence

model training, bias classification performance, and affect the reliability of evaluation metrics.

As shown in Figure (3), the dataset is dominated by IAQ Class 5, which contains 82,261 observations and accounts for nearly half of all records. Classes 1 and 2 are also well represented, with 36,200 and 29,369 observations, respectively. In contrast, Classes 3 and 6 contain substantially fewer samples, with only 6,321 and 41 observations, respectively. Such disparities indicate a pronounced class imbalance, where some IAQ conditions occur much more frequently than others. This imbalance suggests that subsequent analyses should carefully consider the unequal representation of classes, and that appropriate techniques such as class weighting, stratified sampling, or resampling methods may be required when developing machine-learning models.

Overall, Figure 3 provides a concise summary of the global structure of the dataset and serves as a preliminary assessment of class representativeness. The figure establishes the context for the more detailed group-level analysis presented in Figure ??, where the distribution of IAQ classes is examined separately for each monitoring group.

D. Distribution of Indoor Air Quality Classes Across Monitoring Groups

Figure 4 presents the relative distribution of indoor air quality (IAQ) classes for each monitoring group included in the dataset.

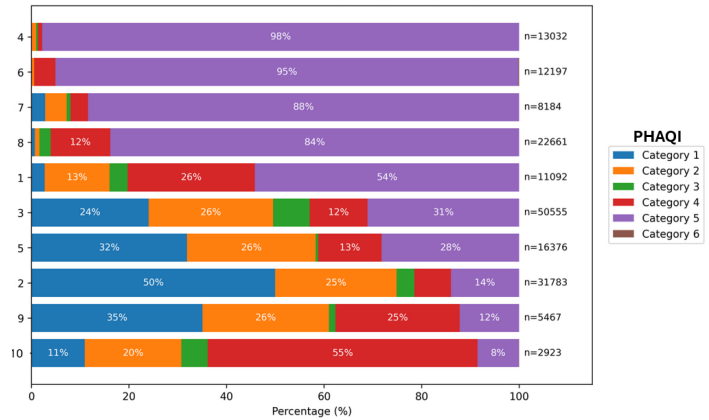


Fig. 4. Distribution of PHAQI by class ID

The figure was constructed from the frequency counts obtained through a pivot-table aggregation of the original observations. For each group, the number of observations belonging to each IAQ class was first calculated and then normalized by the total number of observations within that group. The resulting percentages were displayed using a 100% stacked horizontal bar chart, where each colored segment represents the proportion of a specific IAQ class. The total number of observations available for each group (n) is indicated at the

right side of the corresponding bar to provide information on sample size and data representativeness.

The primary objective of this visualization is to provide a concise overview of the class composition of the dataset and to facilitate the comparison of IAQ distributions among monitoring groups. While absolute frequencies can be strongly influenced by differences in sample size, the use of relative percentages enables a direct comparison of the prevalence of each IAQ class across groups. Consequently, the figure highlights potential imbalances in the dataset and reveals whether certain groups are dominated by a particular IAQ category.

The results indicate substantial variability in class distributions among the monitored groups. Several groups are characterized by a strong predominance of a single IAQ class, whereas others exhibit a more balanced distribution across multiple classes. Such differences suggest that indoor environmental conditions are not uniformly represented throughout the dataset and may reflect variations in building characteristics, occupancy patterns, ventilation strategies, or environmental conditions. Furthermore, the presence of dominant classes in some groups points to a potential class imbalance that should be considered during subsequent statistical analyses or machine-learning model development, as highly represented classes may influence model training and predictive performance.

Overall, Figure 4 provides an informative summary of the dataset structure, allowing the reader to assess data heterogeneity, class balance, and the representativeness of individual monitoring groups before proceeding to more detailed analyses.

LIMITATIONS

The dataset has been collected over less one-year period in an internal area of different schools from San Luis Potosí Metropolitan Area. Consequently, some gaps in the collected data series are possible. Furthermore, the monitoring locations were selected to be representative of a medium-sized classrooms in elementary schools in Mexico. To enhance the representativeness of the dataset across the entire airport environment, future monitoring campaigns should incorporate additional IoT sensing devices in other operational areas and expand the set of monitored parameters to include Total Volatile Organic Compounds (TVOC) as well as outdoor environmental conditions such as temperature and relative humidity.

ETHICS STATEMENT

The authors confirm compliance with the ethical requirements for publication and declare that the present work does not involve human participants, animal experiments, personal data, or data collected from social media platforms.

DATA AVAILABILITY

The original dataset supporting this study is publicly available through the Mendeley Data repository:

<https://doi.org/xxxxxxxxx>

CREDIT AUTHOR STATEMENT

Salvador Ruiz-Correa: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – review & editing, Project administration, Funding acquisition.

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DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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